Think Academy

G1

Term A

M

Think Academy

Secondary School

Lesson L1-L8

Think Academy

NAME: ____________

PREFACE

Welcome to your Think Academy maths course, we're delighted that you've chosen to learn with us!

About Think Academy

Think Academy is part of the NYSE-Ested TAL Education Group, trusted by over 30 million students and their families across the globe.

Over the past 18 years, Think Academy's expertise has focused on curriculum development and maths tuition, supporting students to excel in primary school level mathematics.

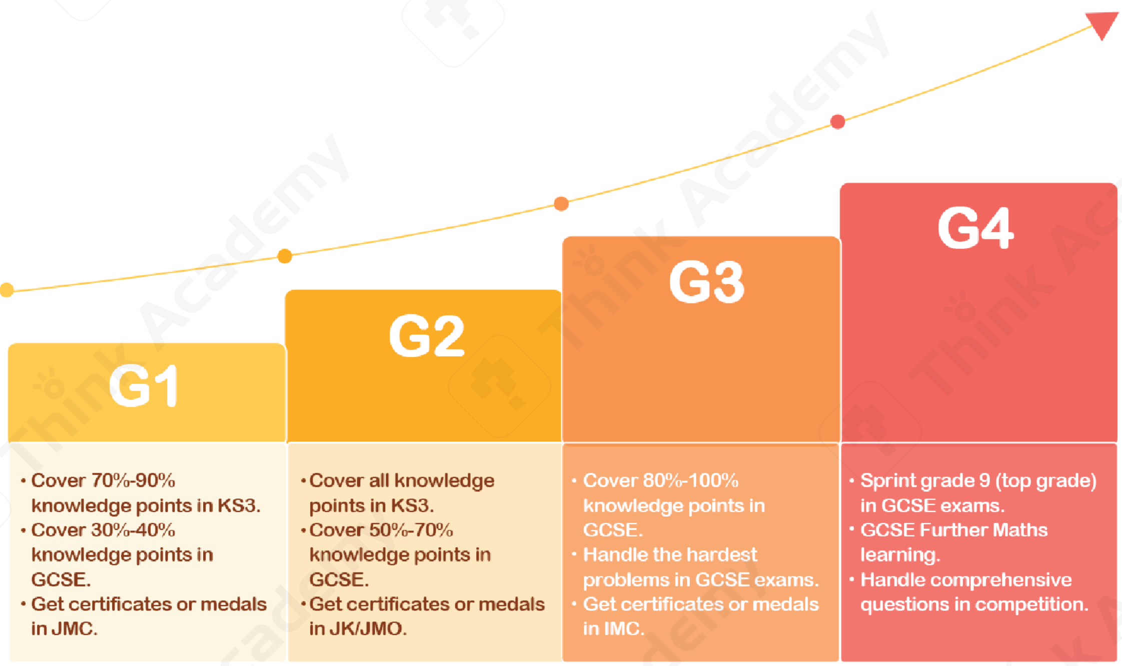
Our bespoke online maths courses harness the power of technology to ensure students achieve their full potential at junior middle school and throughout their GCSE journey and solo challenges of UKMT.

Throughout this online maths course, you and your child will experience a fully guided learning process, led by their tutor with additional support from the tutor's assistant.

This course will introduce your child to new and challenging mathematical concepts. The tutor will encourage your child to break boundaries by using the problem-solving techniques and fast calculation skills required to achieve top grades in the GCSE examination and get an invitation for a further competition of UKMT.

When it's time for class, open the Think Academy Classroom app to get started.

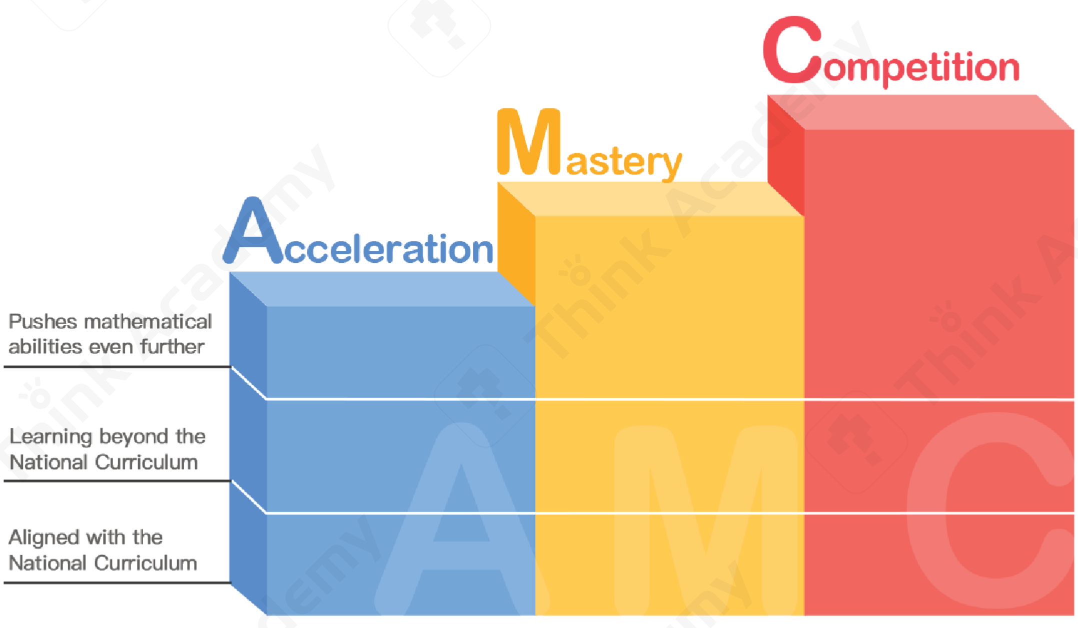
Our curriculum



Think Academy's curriculum is designed to ensure students develop the necessary skills required for success in the GCSE exam and solo challenges of UKMT. That is why our curriculum centres around the 9 modules consistently used in GCSE exams and the competition.

Our curriculum

Level introduction



	Acceleration	Mastery	Competition
Suitable for	GCSE candidates targeting grade 9 (top grade)	GCSE candidates targeting grade 9 (top grade). Prepare for UKMT competition or GCSE Further Maths.	GCSE candidates targeting grade 9 (top grade). Prepare for GCSE Further Maths. Purpose to get certificates or medals in the UKMT competition.

Learning Guide

Before Class

Let's Get Ready

Think about the skills and topics you have already learned that will be useful for this lesson!

In Class

Learn and Discover

Build on what you know by learning more about the topic, and discover new ways to answer questions!

Exploration

Your teacher will help you to understand the topic, exploring smart ways to answer difficult questions!

Practice

Now you can try to answer some questions using your new skills!

Challenge

Stretch yourself by using your new skills to answer some more difficult questions!

After Class

Exercise book

Some exercises from basic level to challenging level to consolidate what you have learnt in each lesson!

Stage Test

We provide a stage test every four lessons, to help you identify your strengths and weaknesses so that tutors can provide targeted instruction to help you improve. It also gives you an opportunity to understand your own progress over time!

Welcome to the Think Academy Team!

Hi! My name's Pip. I'm going to be your guide as we go on an amazing maths adventure together!

And my name's Bud! I think maths can be fun for everyone - let me show you why!

There are lots of different things to learn, but don't worry, we'll be with you every step of the way!

We'll teach you some super smart and fun ways to work out the most difficult maths problems…

Congratulations!
You have completed your adventure!

Contents

Lesson 1

Rational Numbers and Number Line

Target Grade

Knowledge Points	Target Grade	Topic Area
Rational number	5	Number
Number line	5	
Additive inverses	3	
Absolute value	4	

Objectives

- ☑ Be able to understand the meaning of rational numbers.
- ☑ Be able to recognise the number line and solve problems using the number line.
- ☑ Be able to understand additive inverses and absolute value and solve equations.

Let's Get Ready

1 We can use positive numbers and negative numbers to represent opposite quantities.

(1) Walking east $3\ \mathrm{m}$ can be written as $+3$

Walking west $5\ \mathrm{m}$ can be written as ______

(2) Earning £10 can be written as $+10$

Spending £2 can be written as ______

(3) If $+2\%$ means increase 2%, then -6% means ______.

A. increase 2% B. increase 6% C. decrease 2% D. decrease 6%

2 The standard mass of a bag of flour is $50 \pm 0.2\ \mathrm{kg}$, and now ten bags of flour are randomly selected for testing. The results are shown in the table below:

No.	1	2	3	4	5	6	7	8	9	10
Mass (kg)	50	50.1	49.9	50.1	49.7	50.1	50	50	49.9	49.95

The number of defective bags of flour is ______.

A. 1 bag B. 2 bags C. 3 bags D. 4 bags

3 In a long jump competition, the required standard is $2.00\ \mathrm{m}$.

Fei jumps $2.12\ \mathrm{m}$, denoted by $+0.12\ \mathrm{m}$.

Ye jumps $1.95\ \mathrm{m}$, denoted by ______ m.

Ping's result is denoted by $0\ \mathrm{m}$, in fact, the distance of his long jump is ______ m.

In Class

Rational numbers

Notes

1.Definition of positive and negative numbers

(1) Positive numbers: numbers ____________ zero.

Example: $3,\ 4.5,\ \frac{1}{6},\ 18\%$

Can also be written with a '+' in front, like $+3,\ +4.5,\ +\frac{1}{6},\ +18\%$

The word 'positive' can be shortened to '+ve'.

(2) Negative numbers: numbers ___________ zero.

A negative number is written with a minus sign in front.

Example: $-3,\ -5.2,\ -3\frac{1}{2},\ -7.2\%$

The word 'negative' can be shortened to '−ve'.

(3) Zero: neither positive nor negative.

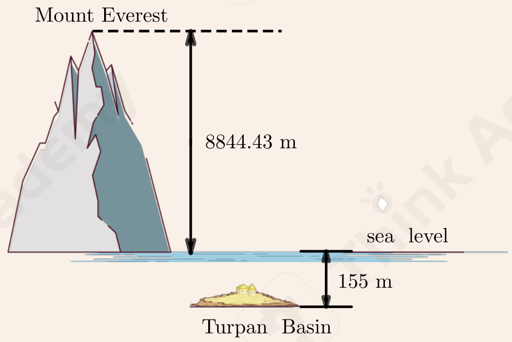
Tips: The plus sign can be removed for positive numbers, but the minus sign cannot be omitted for negative numbers.

2. Negative numbers in context

(1) Sea level

Sea level is used as a point of reference (zero).

The height of Mount Everest is $+8844.43\ \mathrm{m}$, so what is the height of the Turpan Basin?



(2) Temperature

We define the freezing point of water as zero degrees Celsius ($0^{\circ}C$). If it gets colder we need negative temperatures. So $20^{\circ}C$ below zero is ______ $^{\circ}C$

3. Integers

If we include negative whole numbers and zero with positive whole numbers, we get the set of **integers**.

Integers: $\{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$

The set of integers

includes ____

extending indefinitely in both directions.

4. Fractions

A fraction represents how many equal parts of a whole there are.

$$\frac{\text{Numerator}}{\text{Denominator}}$$

We call the top number the Numerator. It is the number of parts we have.

We call the bottom number the Denominator. It is the number of parts the whole is divided into.

5. Decimals

$$Decimals\begin{cases} Terminating\ Decimals \rightarrow Fractions \\ Infinite\ Decimals\begin{cases} Recurring\ Decimals \rightarrow Fractions \\ Infinite\ Non-Recurring\ Decimals\ (Irrational\ numbers) \end{cases} \end{cases}$$

6. Rational Numbers

Rational numbers are numbers that can be represented in the form of $\frac{p}{q}$ where p and q are integers and $q \neq 0$

Learn and Discover

(1) Explain why the following numbers are rational numbers:
$0, 8, -3, 0.12, 7.509, 0.\dot{6}, 2.1\dot{5}$

(2) Do you think π is a rational number?
The square of b is 5, do you think b is a rational number?

Notes

Rational numbers include _________ and _________ (including ___________ decimals and _________ decimals).

Exploration 1

1 Choose from the numbers below.

$-\frac{1}{3}$, 11.11111 , $95.\dot{5}\dot{7}$, $\frac{\pi}{8}$, 0 , $+2004$, -2 , 1.12122122212222 , $-\frac{1}{11}$, π

negative numbers: ______________________

rational numbers: ______________________

2 Investigate whether the statements are true. Give examples and/or counterexamples to justify your answers.

(a) Decimals are rational numbers.

(b) Rational numbers are integers.

(c) Terminating decimals are rational numbers.

(d) The sum of two recurring decimals is a rational number.

(e) Zero is a negative rational number.

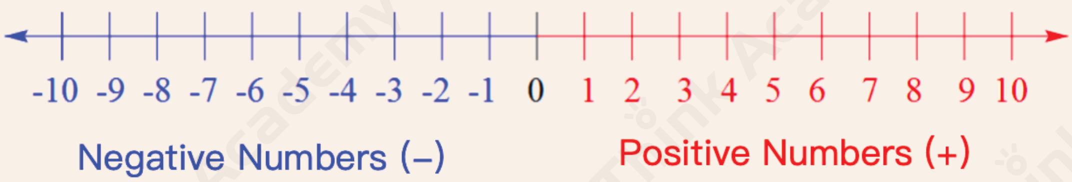
(f) The square of π is a rational number.

Number Line

Notes

1. The definition of the number line

A horizontal line used to represent real numbers as points in their relative positions.



(The line extends infinitely in both directions.)

2. Using the number line to show relations between numbers

A number on the **left** is **less than** a number on the right.

Examples:

• 5 is less than 8

• -1 is less than 1

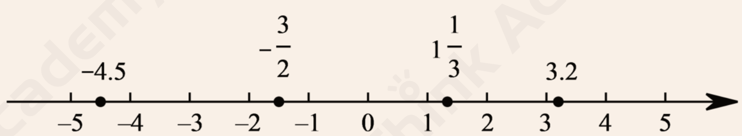
• -8 is less than -5

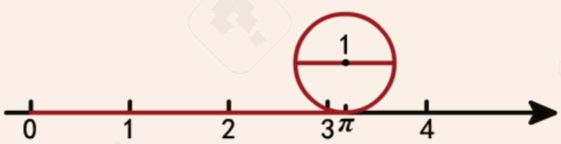
So we can make the following conclusions:

positive numbers > 0, negative numbers < 0, positive numbers $>$ negative numbers

3. The relationship between rational numbers and the number line

All the rational numbers can be represented by points on a number line, but not every point on the number line represent rational numbers. We can find a point on the number line to represent π, but π is an irrational number.





Exploration 2

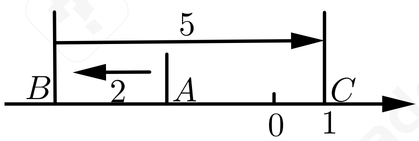
On the number line, there are ______ integers between the two points which represent -200.1 and 200.9 respectively.

Learn and Discover

The point A on the number line represents -5. If you move 6 units to the right along the number line from the point A, you will reach the point B, and the number represented by B is ______

Exploration 3

As shown in the figure, point A on the number line moves 2 units to the left to reach point B, and then moves 5 units to the right to reach point C. If the number represented by point C is 1, the number represented by point A is ______.



A. 7

B. 3

C. -3

D. -2

Challenge

A moving point Q starts from the origin O and moves left and right at a rate of 2 units per second on the number line.

It moves 1 unit to the right, followed by 2 units to the left, 3 units to the right, 4 units to the left, then 5 units to the right, $\cdots$, and so on.

(1) When the moving point Q has moved for 3 seconds, find the number represented by Q on the number line at this moment.

(2) When the moving point Q reaches the point which represents 10 on the number line for the first time, find the time taken for the movement of Q.

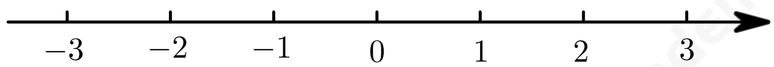
Exploration 4

A bug first climbs 2 units to the right and then 6 units to the left on the number line. The end is just 2 units away from the origin of the number line. So the number represented by the starting position of the bug is ______

Additive Inverses

Learn and Discover

Use the number line to solve the question below:



Walking $2\ \mathrm{km}$ to the east can be recorded as $+2$, and walking $2\ \mathrm{km}$ to the west can be recorded as -2

Pip first walked $2\ \mathrm{km}$ to the east and then $2\ \mathrm{km}$ to the west, where will he stop?

We can express this by the equation $2+(-2)=0$

Using the same method, can you find the relationship between 2.5 and -2.5, 1 and -1?

Notes

1. The Definition of Additive Inverses

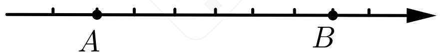
Additive inverses are two numbers whose sum is zero.

2. Property of Additive Inverses

On the number line, the corresponding points of additive inverses are **on opposite sides of the origin** and **at the same distance** from the origin.

Exploration 5

As shown on the number line with a unit length of 1, the two numbers represented by points A and B are additive inverses, so the number represented by point A is ______.



A. 2 B. -2 C. 3 D. -3

Learn and Discover

Find the additive inverses of the numbers below:

(1)

original number	3	0	2.5	a
additive inverses				

(2)

original number	-6	-2	-0.9	$-b$
additive inverses				

(3)

original number	-8	$-(-8)$	$-(+8)$	$+(-8)$
additive inverses				

Notes

1. Find the Additive Inverses of Numbers

To find the additive inverse of a number, we need to place ________________ before the number.

2. Simplification

$+(+a)=a, -(-a)=a$ (Two negatives make a positive.)

$-(+a)=-a, +(-a)=-a$

Exploration 6

1 Which group of the following numbers are additive inverses?

A. $-(+5)$ and $+(-5)$
B. $-(-5)$ and $+(-5)$
C. $-(+5)$ and -5
D. $+(-5)$ and -5

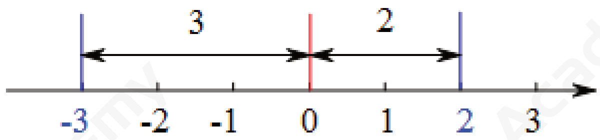
2 If a number moves 8 units to the right to get its additive inverse on the number line, then the number is ______.

A. 4
B. -4
C. 8
D. -8

Absolute Value

Learn and Discover

The absolute value of a number is its distance from zero.



For example, -3 is 3 away from zero, and 2 is 2 away from zero.
So the absolute value of -3 is 3, and the absolute value of 2 is 2
The symbol for absolute value is a bar | on each side of the number.
$|-3| = 3$, $|2| = 2$
Find the absolute values of the following numbers:

original number	-9	2.5	0	-156	2000
absolute value					

Notes

Find the Absolute Value:

- The absolute value of a positive number is itself.
- The absolute value of 0 is 0
- The absolute value of a negative number is its additive inverse.

So when a number is positive or zero we don't change anything; when it is negative we change it to positive using additive inverses.

Exploration 7

Find all the negative integers whose absolute values are greater than 2.4 and smaller than 7.1

Exploration 8

Compare the two numbers with "< ", ">" or "=".

$|-5|$ ______ $|-7|$

$-|6|$ ______ $|-4|$

Learn and Discover

Given that $|a| + |b| = 0$, $a =$ ______, $b =$ ______

Notes

The absolute value of a number is always greater than or equal to 0

It cannot be negative.

If $|a| + |b| = 0$, then $a = 0$ and $b = 0$

★There are similar applications in squares: $a^2 + b^2 = 0$

Exploration 9

Given that $|x - 2| + |2y - 5| = 0$, find the value of $x + y$.

Notes

If the absolute value of a number is positive, we can have two answers: additive inverses.

That means: if $|a| = b(b > 0)$, then $a = b$ or $a = -b$.

Exploration 10

Given that $|x| = 3$, $|y| = 4$, and $x > y$, then the value of y is ______

Challenge

(a) Given that $|x + 3| = 5$

Find all the possible values of x.

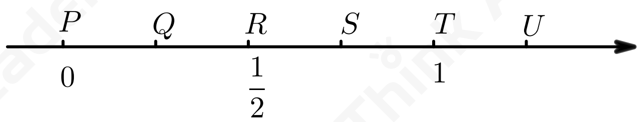
(b) Given that $|2x + 4| = |x - 7|$

Find all the possible values of x.

After Class

Basic Exercises

1 On the number line below, the fraction $\frac{3}{8}$ lies between ______.



A. P and Q B. Q and R C. R and S D. S and T E. T and U

2 On the number line, there are ______ integers between the two points which represent -18.5 and 18.5 respectively.

3 The point representing -1 on the number line moves 4 units to the right and then 5 units to the left. The corresponding number of the endpoint is ______

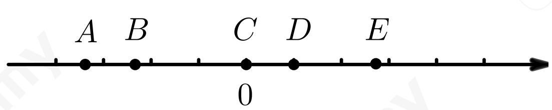
4 An ant starts from the point A on the number line, moves 7 units to the right, and then 4 units to the left. The end point is exactly the origin. The number represented by point A is ______

5 On the number line, the number represented by point A is 8
Starting from point A, point P moves to the right at a speed of 2 units per second. If it reaches point B after 3 seconds, the number represented by point B is ______

6 Which is the smallest of 0, -2, $-\frac{1}{2}$ and 1?

A. 1 B. $-\frac{1}{2}$ C. -2 D. 0

7 As shown on the number line, there are five numbers represented by A, B, C, D and E
Which number has the largest absolute value?



A. A B. B C. C D. D E. E

8 The number of all integers with absolute values greater than 2 and less than 5 is ______.

A. 2 B. 3 C. 4 D. 5

9 (a) Given that $|x-2|+|y-3|=0$, $xy=$ ______

(b) Given that $|8-2x|+|y|=0$, $xy=$ ______

10 Compare the two numbers with $>$, $<$ or $=$.

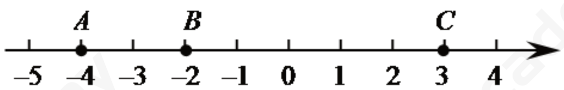
(1) -4 ______ $-|-3|$

(2) $-\frac{5}{6}$ ______ $-\frac{3}{4}$

(3) $-\frac{3}{11}$ ______ -0.273

Extensive Exercises

1 As shown in the figure, there are three points A, B and C on the number line, which represent -4, -2 and 3 respectively.
Answer the questions below:

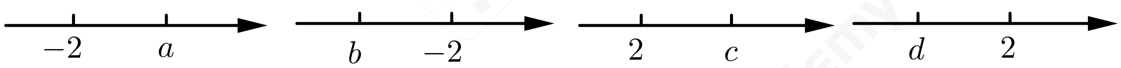


(1) If the point B moves 3 units to the left, the smallest number represented by the three points is _____

(2) To make the distance between C and B equal to the distance between A and B, how many units should point C be moved to the left?

Answer: _____ or _____

2 Which of the following numbers must have an absolute value larger than 2?



A. a and c B. b and c C. a and d D. b and d

3 Given that $|5a-5|$ and $|22-2b|$ are additive inverses, work out the value of $|b-a|$.

Answer: _____

Extensive Challenge

1 A flea jumps on a number line.

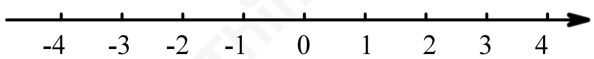
Starting from the position of $+18$, it jumps right for the first time by 1 unit, then left for the second time by 2 units, right for the third time by 3 units, left for the fourth time by 4 units, right for the fifth time by 5 units, and left for the sixth time by 6 units $\cdots$.

When it jumps for 100 times, calculate the number represented by the landing point.

A. -32 B. 32 C. -50 D. 50

2 $|3-1|$ represents the absolute value of the difference between 3 and 1, which can also be understood as the distance between the two points corresponding to 3 and 1 on the number line.

$|3+1|$ can be regarded as $|3-(-1)|$, which represents the absolute value of the difference between 3 and -1, which can also be understood as the distance between the two points corresponding to 3 and -1 on the number line.



(1) $|3-(-1)| =$ ______

(2) Use the number line to solve the questions below:

① if $|x-1| = 3$, $x =$ ______

② if $|x-(-2)| = 5$, $x =$ ______

③ if $|x+2| = 4$, $x =$ ______

Lesson 2

Addition and Subtraction of Rational Numbers

Target Grade

Knowledge Points	Target Grade	Topic Area
Addition of rational numbers	3	Number
Subtraction of rational numbers	3	
Mixed operations involving addition and subtraction of rational numbers	PK: difficulty 3	

Objectives

- Be able to apply addition and subtraction of rational numbers.
- Be able to apply mixed operation of addition and subtraction of rational numbers.

Let's Get Ready

1 Work out each of the following.

(1) $261 + 24 =$

(2) $341 - 52 =$

2 Work out each of the following.

(1) $4.25 + 16 =$

(2) $1.3 - 0.9 =$

(3) $8.12 + 4.33 =$

(4) $16.3 - 8.7 =$

3 Work out each of the following.

(1) $\frac{5}{9} + \frac{2}{9} =$

(2) $5\frac{3}{4} + \frac{1}{8} =$

(3) $3\frac{5}{6} - 1\frac{1}{4} =$

(4) $7 - 4\frac{3}{7} =$

In Class

Learn and Discover

1 We can use counters to represent numbers.

1 1 1 $= 3$ −1 −1 −1 $= -3$

1 1 1 1 1 $= 5$ −1 −1 −1 −1 −1 $= -5$

We can also use counters to perform the calculations.

1 1 1

1 1 1 1 1

$3 + 5 =$ ______

−1 −1 −1

−1 −1 −1 −1 −1

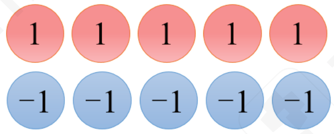
$-3 + -5 =$ ______

Think about how to add two numbers with the same sign and work out each of the following.

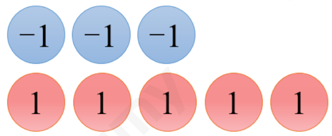
$12 + 8 =$ ______ $9 + 7 =$ ______

$-10 + -6 =$ ______ $-15 + -12 =$ ______

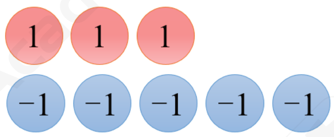
2 Use counters to add a positive number and a negative number.



$5 + -5 =$ _____



$-3 + 5 =$ _____



$3 + -5 =$ _____

Think about how to add two numbers with different signs and work out each of the following.

$-12 + 8 =$ _____ $\quad$ $9 + -7 =$ _____

$10 + -15 =$ _____ $\quad$ $-32 + 11 =$ _____

Notes

Use the fact that a number plus its additive inverse is zero to simplify calculations.

e.g. $-3 + 5 = -3 + 3 + 2 = 2$

e.g. $3 + -5 = 3 + -3 + -2 = -2$

Exploration 1

Work out each of the following.

(1) $(-1.6)+(+2.7)=$ ______

(2) $5\frac{1}{2}+\left(-3\frac{1}{3}\right)=$ ______

(3) $(-5.1)+1\frac{2}{5}=$ ______

(4) $(-6.3)+\left(-2\frac{1}{6}\right)=$ ______

Exploration 2

Jenny had to add 26 to a certain number. Instead, she subtracted 26 and obtained -14

What number should she have obtained?

Exploration 3

Work out each of the following.

(1) $8 + 18 + (-6) + (-5)$

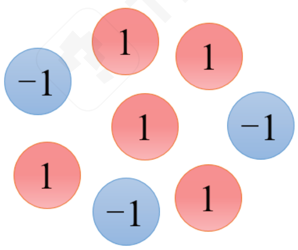
(2) $-3.5 + 7.8 + (-2) + 3.5$

(3) $\frac{3}{7} + \frac{2}{25} + \left(-\frac{3}{7}\right)$

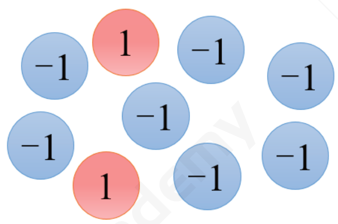
(4) $\frac{1}{2} + 3.5 + (-0.5) + 2 + (-10)$

Learn and Discover

1 (a) Find the totals of these sets of counters.



$=$ ______



$=$ ______

(b) How does the total change if you remove a red counter?

(c) How does the total change if you remove a blue counter?

(d) Work out each of the following.

$30 - 1 =$ ______ $\quad 30 - (-1) =$ ______

$(-10) - 6 =$ ______ $\quad (-10) - (-6) =$ ______

2 We know that $-(-6) = +6$, we can use it to perform the calculations, for example:

$12 - (-6) = 12 + 6 = 18$

$(-28) - (-6) = -28 + 6 = -22$

Work out each of the following.

$60 - (-50) =$ ______

$(-50) - (-30) =$ ______

Notes

When adding and subtracting with negative numbers:

Step 1: Simplify the signs in the calculation.

Two negatives make a positive. e.g. $2 - (-1) = 2 + 1$

When simplifying adjacent signs, a minus followed by a plus gives a minus.

e.g. $2 + (-1) = 2 - 1, 2 - (+1) = 2 - 1$

Step 2: Convert subtraction to addition if necessary.

Exploration 4

Work out each of the following.

(1) $-5.7 - (+2.9) =$ ______

(2) $3\frac{2}{3} - \left(-\frac{5}{6}\right) =$ ______

(3) $3.5 - 5\frac{1}{4} =$ ______

(4) $-4.6 - \left(+2\frac{3}{5}\right) =$ ______

Practice

Work out each of the following.

(1) $-\frac{2}{5} - \frac{1}{6} =$ ______

(2) $-0.75 - (-\frac{7}{12}) =$ ______

Exploration 5

Work out each of the following.

(1) $-12 - (-18) + (-7) - 15$

(2) $-20 + (+3) - (-5) - (+7)$

(3) $20 + (-14) - (-18) - 13$

(4) $-7 + (-5) - (-6) - (+2)$

Exploration 6

1. What is $2010 + (+2010) + (-2010) - (+2010) - (-2010)$?

A. 0 B. 2010 C. 4020 D. 6030 E. 8040

2. What is the value of $2 - (-2 - 2) - (-2 - (-2 - 2))$?

A. 0 B. 2 C. 4 D. 6 E. 8

Challenge

Calculate $|-1| + (-3) + |-5| + (-7) + \cdots + |-97| + (-99)$

After Class

Basic Exercises

1 Work out each of the following.

(1) $-81+(-29)=$ ______

(2) $-17+21=$ ______

(3) $0+(-45)=$ ______

(4) $-13+13=$ ______

(5) $-25+50=$ ______

2 Work out each of the following.

(1) $-35+(-2.3)$

(2) $-1.2+1.8$

(3) $-18.2+(-9.4)$

3 Work out each of the following.

(1) $-17-(-85)=$ ______

(2) $-54-(+19)=$ ______

(3) $23-(-112)=$ ______

(4) $185-203=$ ______

4 Work out each of the following.

(1) $13.4-(-6.7)$

(2) $-20.4-(+5.3)$

(3) $-36-(-2.8)$

5 Work out each of the following.

(1) $-2\frac{1}{5}+(-1\frac{3}{5})$

(2) $-4\frac{2}{3}-1\frac{1}{9}$

(3) $2\frac{2}{3}-3\frac{3}{4}$

(4) $3-(-6\frac{3}{16})$

6 Work out each of the following.

(1) $-\frac{2}{5}-(-\frac{2}{7})$

(2) $4\frac{1}{4}-(-2\frac{5}{6})$

(3) $-4\frac{1}{3}+7\frac{2}{5}$

(4) $2\frac{5}{6}+(-\frac{1}{8})$

7 Work out each of the following.

(a) $-4.6+19.7+4.6$

(b) $-8+(-9)+(-10)+20$

(c) $29+(-7)+16+(-30)$

(d) $4.12+(-9.7)+(-4.12)+2.4$

8 Work out each of the following.

(1) $13 + (-12) + 17 + (-18) =$ ______

(2) $-18.75 + 6.25 + (-3.25) + 18.75 =$ ______

(3) $\frac{1}{2} - 2\frac{1}{4} - 3\frac{1}{2} + 2.25 =$ ______

9 A police officer rode a motorcycle to patrol a north-south road. One day, he started from the sentry box and stopped at place A at night. The driving records of the day are as follows (unit: km), the positive numbers mean driving north and the negative numbers mean driving south:

$+10$, -9 , $+7$, -15 , $+6$, $+4$, -14 , -2

(1) In which direction is place A from the sentry box? How far is it from the sentry box?

(2) If the motorcycle consumes 0.05 litres of fuel per kilometre, how many litres of fuel will it consume on that day?

10 Instead of subtracting 8 from a number, Wilson added 8 to that number and obtained -20

What number should he have obtained?

Answer: ______

Extensive Exercises

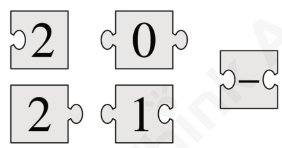
1 Little Lucas invented his own way to write down negative numbers before he learned the usual way with the minus sign in front.

Counting backwards, he would write: $3, 2, 1, 0, 00, 000, 0000, \cdots$.

What is the result of $000 + 0000$ in his notation?

A. 1 B. 00000 C. 000000 D. 0000000 E. 00000000

2 When the five pieces shown are fitted together correctly, the result is a rectangle with a calculation written on it. What is the answer to this calculation?



A. -100 B. -8 C. -1 D. 199 E. 208

3 Work out each of the following.

(1) $1-(-4+(-2))-(8-(-5-7))=$ ______

(2) $1-(2-(3-(4-5)))=$ ______

Extensive Challenge

1 Calculate $|-1| + (-5) + |-9| + (-13) + \cdots + |-73| + (-77) =$ ______

2 If p is a positive integer and q is a negative integer, which of the following is greatest?

A. $p - q$ B. $q - p$ C. $p + q$ D. $-p - q$

Lesson 3

Multiplication and Division of Rational Numbers

Target Grade

Knowledge Points	Target Grade	Topic Area
Multiplication of rational numbers	3	Number
Division of rational numbers	3	
Mixed operation of multiplication and division of rational numbers	JMO: difficulty 4	

Objectives

- Be able to apply multiplication and division of rational numbers.
- Be able to apply mixed operations of multiplication and division of rational numbers.

Let's Get Ready

1 Work out each of the following.

(1) $\frac{4}{9} \times 36$

(2) $\frac{7}{4} \times \frac{8}{21}$

(3) $\frac{8}{7} \times \frac{21}{16}$

(4) $\frac{14}{5} \times \frac{15}{7}$

2 Work out each of the following.

(1) $\frac{3}{5} \div 9$

(2) $\frac{6}{5} \div \frac{2}{15}$

(3) $\frac{9}{4} \div \frac{27}{8}$

(4) $\frac{12}{5} \div \frac{3}{4}$

3 Work out each of the following.

(1) $\frac{4}{5} \times 1\frac{7}{8}$

(2) $4\frac{1}{3} \times \frac{3}{7}$

(3) $2\frac{4}{13} \div 3\frac{3}{4}$

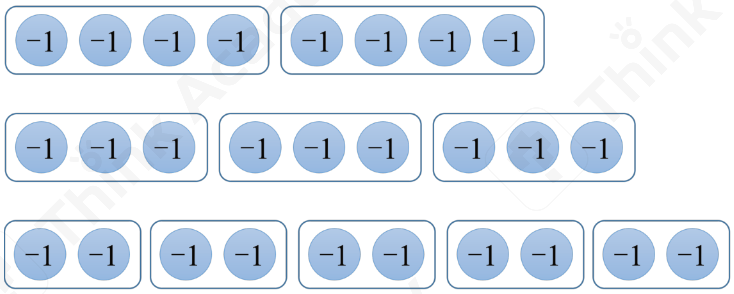
(4) $1\frac{4}{7} \div 2\frac{5}{14}$

In Class

Multiplication of Rational Numbers

Learn and Discover

1 Complete the equations:



$-4 \times 2 =$ _____

$-3 \times$ _____ $=$ _____

$-2 \times$ _____ $=$ _____

Work out each of the following.

$-7 \times 8 =$ _____ $-20 \times 3 =$ _____

$12 \times (-5) =$ _____ $6 \times (-11) =$ _____

2 Work out each of the following.

$-5 \times 3 =$ _____

$-5 \times 2 =$ _____

$-5 \times 1 =$ _____

$-5 \times 0 =$ _____

$-5 \times (-1) =$ _____

$-5 \times (-2) =$ _____

$-5 \times (-3) =$ _____

What happens when you multiply two negative numbers?

Work out each of the following.

$-6 \times (-3) =$ _____ $-10 \times (-8) =$ _____

$-2 \times (-3) \times (-4) =$ _____ $-1 \times (-2) \times (-3) \times (-4) =$ _____

Notes

Multiplication rules of rational numbers:

(1) **Determining the sign of a product**: When multiplying several non-zero numbers, the sign of the product is determined by the number of negative factors.

When the number of negative factors is odd, the product is ________.

When the number of negative factors is even, the product is ________.

(2) When multiplying several numbers, if one of the factors is 0, the product is 0

Exploration 1

1 Work out each of the following.

(1) $+12 \times \left(+\frac{2}{3}\right) =$ ______

(2) $-\frac{5}{6} \times \frac{12}{7} =$ ______

(3) $2\frac{1}{5} \times \left(-\frac{7}{11}\right) =$ ______

(4) $-2\frac{5}{6} \times \left(-2\frac{2}{17}\right) =$ ______

2 Work out each of the following.

(1) $-0.4 \times \left(-1\frac{1}{3}\right) \times \left(-2\frac{1}{2}\right)$

(2) $-3 \times \left(-1\frac{4}{5}\right) \times \left(-1\frac{1}{9}\right) \times \left(+5\frac{1}{2}\right) \times \frac{3}{11}$

Exploration 2

Kanga wants to multiply three different numbers from the following list: $-5, -3, -1, 2, 4, 6$
What is the smallest result she could obtain?

A. -200 B. -120 C. -90 D. -48 E. -15

Exploration 3

Four different integers p, q, r and s satisfy the equation $(5-p)(5-q)(5-r)(5-s) = 25$
What is the value of $p+q+r+s$?

A. 15 B. 20 C. 25 D. 30 E. 35

Challenge

The product of five different integers is 12
Work out the largest integer.

Exploration 4

1 Two numbers have a sum of 2 and a product of -960
Find the positive difference between the two numbers.

2 Three different integers have a sum of 1 and a product of 36
Work out these three different integers.

Challenge

A list of integers has a sum of 2018, a product of 2018, and includes the number 2018
Which of the following could be the number of integers in the list?

A. 2016 B. 2017 C. 2018 D. 2019 E. 2020

Division of Rational Numbers

Learn and Discover

If we know $-5 \times 3 = -15$, we also know:

$-15 \div 3 =$ ______ $-15 \div (-5) =$ ______

If we know $-4 \times -6 = 24$, we also know:

$24 \div (-4) =$ ______ $24 \div (-6) =$ ______

Work out each of the following.

$-42 \div (-7) =$ ______ $-25 \div (-5) =$ ______

$30 \div (-3) =$ ______ $48 \div (-4) =$ ______

Warm Up

If the product of two numbers is 1, the two numbers are reciprocals.

For example: $\frac{2}{3} \times \frac{3}{2} = 1$, thus the reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$ (or the reciprocal of $\frac{3}{2}$ is $\frac{2}{3}$).

Write down the reciprocal of each of the following numbers in the blanks.

The reciprocal of $\frac{3}{7}$ is ______ The reciprocal of $\frac{1}{5}$ is ______

The reciprocal of 12 is ______ The reciprocal of $2\frac{3}{4}$ is ______

The reciprocal of $-\frac{7}{9}$ is ______

Notes

Division rules of rational numbers:

(1) Dividing by a non-zero number is the same as multiplying by its reciprocal.

(2) Zero divided by any non-zero number is zero.

Exploration 5

Work out each of the following.

(1) $-0.75 \div \frac{5}{4}$

(2) $-\frac{2}{7} \div (-4)$

(3) $-25 \div \frac{1}{5}$

(4) $13 \div \left(-\frac{1}{13}\right)$

Practice

Work out each of the following.

(1) $2 \div \left(-\frac{1}{2}\right) =$ ______

(2) $9 \div \left(-\frac{3}{5}\right) =$ ______

(3) $-2\frac{9}{11} \div 31 =$ ______

(4) $-4 \div (-0.25) =$ ______

Exploration 6

Work out each of the following.

(1) $-2\frac{2}{3} \div (-4) \div (-\frac{2}{5})$

(2) $-54 \times 2\frac{1}{4} \div \left(-4\frac{1}{2}\right) \times \frac{2}{9}$

After Class

Basic Exercises

1 Work out each of the following.

(1) $8\times(-7)=$ ______

(2) $-28\div(-4)=$ ______

(3) $3\times(-5)\times(-10)=$ ______

2 Work out each of the following.

(1) $\frac{2}{3}\times\left(-\frac{9}{8}\right)$

(2) $-5\times\left(-\frac{11}{30}\right)$

(3) $-2\frac{2}{3}\times\left(-\frac{9}{8}\right)$

(4) $-2.5\times\frac{11}{15}$

3 Work out each of the following.

(1) $-\frac{4}{5}\div\frac{3}{20}$

(2) $-\frac{5}{9}\div(-\frac{5}{6})$

(3) $\frac{9}{14}\div(-3)$

(4) $-30\div\frac{5}{9}$

(5) $\frac{12}{5}\div(-\frac{3}{10})$

(6) $-\frac{8}{21}\div(-\frac{12}{7})$

4 Work out each of the following.

(1) $2 \div (-\frac{3}{2})$

(2) $-\frac{1}{3} \times (-\frac{3}{4})$

(3) $-\frac{4}{7} \div \frac{7}{4}$

(4) $-\frac{2}{3} \times (-2\frac{1}{2})$

(5) $-3\frac{1}{2} \div \frac{7}{3}$

(6) $-6\frac{1}{4} \times (-3\frac{1}{5})$

5 Work out each of the following.

(1) $-2 \times 9 \times (-8) =$ ______

(2) $-5 \times (-3) \times 4 \times (-2) =$ ______

(3) $-3 \times (-3) \times (-3) \times (-3) =$ ______

(4) $-5 \times 8 \times (-7) \times (-0.25) =$ ______

6 Work out $-\frac{3}{4} \times (-\frac{1}{2}) \div (-2\frac{1}{4})$

7 Work out each of the following.

(1) $-\frac{5}{12} \times \frac{8}{15} \div \left(-\frac{3}{2}\right)$

(2) $-3 \times \left(-\frac{5}{6}\right) \div \left(-1\frac{1}{4}\right)$

(3) $-0.25 \div \left(-\frac{3}{7}\right) \times \frac{4}{3}$

8 Work out each of the following.

(1) $2\frac{5}{6} \times \left(-1\frac{1}{2}\right) \times \frac{6}{17} \div \left(-\frac{2}{3}\right)$

(2) $\frac{1}{6} \times (-6) \div \left(-\frac{1}{7}\right) \times 7$

(3) $-16 \div \frac{1}{3} \times 3 \times \left(-1\frac{7}{8}\right)$

9 Calculate $-72 \times 2\frac{1}{4} \times \left(-\frac{4}{9}\right) \div \left(-3\frac{3}{5}\right) =$ ________

10 Calculate $-5 \div \left(-1\frac{2}{7}\right) \times \frac{4}{5} \times \left(-2\frac{1}{4}\right) \div 7 =$ ________

Extensive Exercises

1. Four different positive integers p, q, r, s satisfy the equation
$(9-p)(9-q)(9-r)(9-s) = 9$
What is the value of $p+q+r+s$?
A. 20 B. 24 C. 28 D. 32 E. 36

2. Two numbers multiply together to make -24
They add together to make 10
These two numbers are _____ and _____

3. Three different integers have a sum of -1 and a product of 24
These three different integers are _____, _____ and _____

Extensive Challenge

1 Given that a, b, c, d, e and f are different integers and $abcdef = -36$

$a + b + c + d + e + f =$ ______

2 Given that $a = 1 \div 2 \div 3 \div 4$

$b = 1 \div (2 \div 3 \div 4)$

$c = 1 \div (2 \div 3) \div 4$

$d = 1 \div 2 \div (3 \div 4)$

The value of $(b \div a) \div (c \div d)$ is ______

Lesson 4

Powers and Numbers in Standard Form

Target Grade

Knowledge Points	Target Grade	Topic Area
Squares, cubes and roots	3	Number
Powers of rational numbers	5 IMC: difficulty 3	
Numbers in standard form		

Objectives

- Be able to understand squares, cubes and roots of rational numbers.
- Be able to calculate higher powers of positive or negative rational numbers.
- Be able to write numbers in standard form and compare numbers in standard form.

Let's Get Ready

1. Work out each of the following.

$8^2 =$ ______

$15^2 =$ ______

$9^3 =$ ______

$20^3 =$ ______

2. Work out each of the following.

$49 = \_\_\_\_\_\_^2$

$900 = \_\_\_\_\_\_^2$

$64 = \_\_\_\_\_\_^3$

$125 = \_\_\_\_\_\_^3$

3. Work out each of the following.

$3.57 \times 10 =$ ______

$4.639 \times 100 =$ ______

$0.0239 \times 1000 =$ ______

$123.4 \div 10 =$ ______

$12.7 \div 100 =$ ______

$33.45 \div 1000 =$ ______

In Class

Squares, Cubes and Roots

Warm Up

Work out each of the following.

(a) $2^2 =$ ______ $(-2)^2 =$ ______ $5^2 =$ ______ $(-5)^2 =$ ______

(b) $2^3 =$ ______ $(-2)^3 =$ ______ $5^3 =$ ______ $(-5)^3 =$ ______

Learn and Discover

The square root of a number is defined as the number which results in the original number when it is squared.

$3 \times 3 = 9$, so 3 is the square root of 9

$-3 \times (-3) = 9$, so -3 is also the square root of 9

We can write the positive square root of x as $\sqrt{x}$ and the negative square root as $-\sqrt{x}$:

$$\sqrt{9} = 3 \qquad -\sqrt{9} = -3$$

Work out each of the following.

$\sqrt{100} =$ ______ $-\sqrt{0.49} =$ ______

$\sqrt{\frac{4}{9}} =$ ______

Notes

(1) A positive number has two different square roots, which have the same value but opposite signs.
But when we use the **square root symbol** $\sqrt{\ }$, we only give the **positive (or zero) result**, like: $\sqrt{9} = 3$
(2) A negative number doesn't have square roots.

Learn and Discover

The cube root of a number is defined as the number which results in the original number when it is cubed.
$3 \times 3 \times 3 = 27$, so 3 is the cube root of 27
$-3 \times -3 \times (-3) = -27$, so -3 is the cube root of -27
The sign $\sqrt[3]{\ }$ is used as a cube root symbol:

$$\sqrt[3]{27} = 3 \qquad \sqrt[3]{-27} = -3$$

Work out each of the following.

$\sqrt[3]{-8} =$ ______

$\sqrt[3]{0.125} =$ ______

$\sqrt[3]{\dfrac{27}{125}} =$ ______

Notes

(1) A positive number only has one cube root, which is positive.
(2) A negative number only has one cube root, which is negative.

Exploration 1

1 Answer the following questions.

(1) Write down the value of $\sqrt{3.61}$

(2) Find the square root of $\frac{49}{64}$

(3) Find the square roots of 5

(4) Find the square root of $\sqrt{16}$

2 Work out each of the following.

(1) $-\sqrt[3]{\frac{1}{64}} =$ ______

(2) $\sqrt[3]{3\frac{3}{8}} =$ ______

(3) $\sqrt[3]{-0.027} =$ ______

(4) $\sqrt[3]{(-2)^3} =$ ______

Powers of rational numbers

Learn and Discover

Calculate and determine whether the results are positive or negative in each case.

(a) $3^4 =$ ______ $\quad 3^5 =$ ______

(b) $-3^4 =$ ______ $\quad -3^5 =$ ______

(c) $(-3)^4 =$ ______ $\quad (-3)^5 =$ ______

Exploration 2

Work out each of the following.

(1) $-2^3 =$ ______

(2) $\left(-\frac{3}{5}\right)^4 =$ ______

(3) $-\left(\frac{3}{5}\right)^4 =$ ______

(4) $\frac{3^4}{5} =$ ______

Exploration 3

Write these numbers in order of size.

Start with the smallest number.

$(-0.6)^3$, 0.6^0, -0.6^2, $(-0.6)^6$, -0.6

Exploration 4

Given that $|x-3|+(y+2)^2=0$, then $y^x=$ ______

Practice

Given that $(3a+1)^2$ and $(b-1)^2$ are additive inverses, then $a+(-b)^{2018}=$ ______

Numbers in Standard Form

Learn and Discover

1 Work out each of the following.

$10^3 =$ ______
$10^2 =$ ______
$10^1 =$ ______
$10^0 =$ ______
$10^{-1} =$ ______
$10^{-2} =$ ______
$10^{-3} =$ ______

2 Standard form is useful for writing very big or very small numbers in a more convenient way. Any number can be written in standard form exactly like this:

A must always be between 1 and 10 → $A \times 10^n$ ← n can be any integer

For example:

$$421000 = 4.21 \times 10^5$$

$$0.000421 = 4.21 \times 10^{-4}$$

Write the following numbers in standard form.

(a) $52000000 =$ ______
(b) $0.0000325 =$ ______

Notes

1. Standard form

Standard form is a way of writing very large numbers or very small numbers.

$A \times 10^n$: $|A|$ must be greater than or equal to 1 and it also must be less than 10, and n must be an integer.

Examples	2×10^3, 1.01×10^{-4}, -2×10^3
Non-examples	0.3×10^2, 35.1×10^{-4}, $2.4 \times 10^{0.5}$

2. How to Do it

To figure out the power of 10, think "how many places do I move the decimal point?"

(1) When the absolute value of the number is 10 or greater, the decimal point has to move **to the left** to the first non-zero digit, and the power of 10 is ________.

(2) When the absolute value of the number is smaller than 1, the decimal point has to move **to the right** to the first non-zero digit, so the power of 10 is ________.

Tips:

1 million = 1×10^6

1 billion = 1×10^9

Exploration 5

Write the following numbers in standard form.

(a) $6014000000 =$ _____

(b) $0.00000059 =$ _____

(c) 63 million $=$ _____

(d) 17600 billion $=$ _____

(e) 7800 km $=$ _____ cm

(f) 0.05 mm $=$ _____ km

Exploration 6

A satellite orbits the Earth at a speed of 8×10^3 m/s.
Express the distance the satellite travels in one hour in standard form.

Practice

The mass of the Sun is 2×10^{30} kg.
The mass of the largest known star is 315 times the mass of the Sun.
Work out the mass of this star.
Give your answer in kg in standard form.

Exploration 7

Complete these statements using >, < or =.

5.6×10^6 ______ 6×10^5 71000 ______ 7.1×10^5

Challenge

Write these numbers in order of size.

Start with the smallest number.

2.73×10^3, 27.3×10^{-3}, 273×10^2, 0.00273

After Class

Basic Exercises

1 Work out each of the following.

(1) $\sqrt{1.69}$

(2) $-\sqrt{\dfrac{64}{289}}$

(3) $\sqrt[3]{0.064}$

(4) $\sqrt[3]{-\dfrac{27}{125}}$

2 How many statements are true?

i. Negative numbers don't have square roots, but they have cube roots.

ii. The square roots of $\dfrac{4}{9}$ are $\pm\dfrac{2}{3}$

iii. The cube root of $\dfrac{8}{27}$ is $\pm\dfrac{2}{3}$

iv. The cube root of -8 is -2

A. 1 B. 2 C. 3 D. 4

3 The square roots of $\sqrt[3]{64}$ are ______

4 Work out each of the following.

(1) $\left(\frac{2}{3}\right)^3$

(2) $\frac{2^3}{3}$

(3) $\left(-\frac{2}{3}\right)^3$

(4) $-\frac{(-2)^3}{3}$

5 n is negative.
Circle the expression that is **positive**.

A. $n-1$ B. n^2 C. n^3 D. $\frac{1}{n}$

6 Given that $|a+5|+(b-4)^2=0$, work out the value of $(a+b)^{2021}$

A. 1 B. -1 C. 0 D. 2021

7 Among $(-1)^2$, $(-1)^3$, -1^2, $|-1|$, $-(-1)$, $-\frac{1}{-1}$, which of these evaluate to 1?

A. 3 B. 4 C. 5 D. 6

8 Given that $|x+1|+\sqrt{z-2}+(2y-4)^2=0$, then $x+y+z=$ ______

9 (1) Write 0.00562 in standard form.

(2) Write 1.452×10^3 as an ordinary number.

10 (a) Write 75400000 in standard form.

(b) Write 0.000068 in standard form.

Extensive Exercises

1 An integer is written as 8.1555×10^{10}
How many zeros are there in the ordinary number?

A. 4 B. 6 C. 7 D. 10

2 Pierre is given this question.
Pierre's answer is 24.4×10^{7}
Is Pierre correct?
Explain your answer.

> Work out.
>
> 61000×4000
>
> Give your answer in standard form.

3 Write the following numbers in order of size.
Start with the smallest number.
0.038×10^{-6}, 3800×10^{-10}, 380, 3.6×10^{-7}

Extensive Challenge

1 The nearest star to the solar system is called Proxima Centauri, which is about 4.2 light years away from the solar system. A light year is a unit of length in astronomy that measures the distance between celestial bodies. If 1 light year is about $9\ 500\ 000\ 000\ 000$ kilometres, what is the approximate distance from the solar system to Proxima Centauri?

A. 4×10^{13} km B. 4×10^{12} km C. 9.5×10^{13} km D. 9.5×10^{12} km

2 If n is a positive integer, then $(-1)^n + (-1)^{n+1} =$ ______

Stage Test (L1-L4)

1. Work out the number of integers between -25.2 and 25.2 on a number line.

2. The point representing -4 on the number line moves 9 units to the right and then 12 units to the left. Write down the number that represents its final position.

3. A counter starts from point A on the number line, moves 8 units to the right, and then 13 units to the left.
The end point is exactly the origin.
Work out the value represented by point A.

4 Work out each of the following.

(1) $-42+(-3.6)$

(2) $-2.4+3.9$

(3) $-24.6+(-8.7)$

5 Work out each of the following.

(1) $15.2-(-4.8)$

(2) $-18.6-(+7.1)$

(3) $-45-(-3.2)$

6 Work out each of the following.

(1) $-3\frac{2}{7}+\left(-2\frac{4}{7}\right)$

(2) $-5\frac{3}{4}-1\frac{1}{8}$

(3) $3\frac{1}{4}-4\frac{2}{5}$

(4) $4-\left(-5\frac{5}{12}\right)$

7 Work out each of the following.

(1) $-4\times 7\times -5$

(2) $-2\times(-6)\times 5\times(-3)$

(3) $-2\times(-2)\times(-2)\times(-2)\times(-2)$

(4) $-4\times 15\times(-6)\times(-0.5)$

8 Calculate $\left(-\frac{2}{5}\right) \times \left(-\frac{3}{4}\right) \div \left(-1\frac{1}{5}\right)$

Give your answer as a fraction in its simplest form.

9 Work out each of the following.

(1) $\sqrt{1.96} =$ ______

(2) $-\sqrt{\frac{81}{256}} =$ ______

(3) $\sqrt[3]{0.027} =$ ______

(4) $\sqrt[3]{-\frac{64}{125}} =$ ______

10 Find the square roots of $\sqrt[3]{729}$

Lesson 5

Laws of Arithmetic

Target Grade

Knowledge Points	Target Grade	Topic Area
Commutative law, associative law and distributive law	4	Number
Using different laws to flexibly and efficiently solve problems	IMC: difficulty 4	

Objectives

- Be able to apply the commutative, associative and distributive laws to simplify calculations involving decimal or fraction multiplications.
- Be able to use different laws to flexibly and efficiently finish calculation.

Let's Get Ready

1. Calculate $\frac{7}{15} \times \frac{5}{14} \div \frac{1}{18}$

2. To calculate $(-3) \times \left(4 - \frac{1}{2}\right)$, which of the following is correct?

   A. $3 \times 4 \times (-3) \times \left(-\frac{1}{2}\right)$

   B. $(-3) \times 4 - (-3) \times \left(-\frac{1}{2}\right)$

   C. $(-3) \times 4 - (-3) \times \frac{1}{2}$

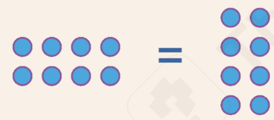
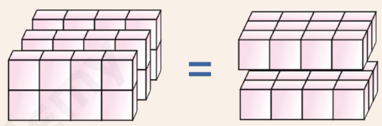
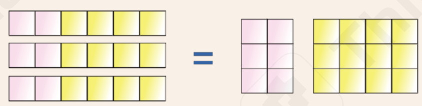
   D. $(-3) \times 4 + 3 \times \left(-\frac{1}{2}\right)$

3. Calculate $\left(\frac{1}{16} - \frac{1}{24}\right) \times (-48) =$ ______

In Class

Commutative Law, Associative Law and Distributive Law

Notes

Description	Symbolically	Example
Commutative law	$a \times b = b \times a$	 2 × 4 = 4 × 2
Associative law	$(a \times b) \times c = a \times (b \times c)$	 (2 × 4) × 3 = 2 × (4 × 3)
Distributive law	$a \times (b + c) = a \times b + a \times c$	 3 × (2+4) = 3×2 + 3×4

Learn and Discover

Which of the following is the correct way to quick calculate 10.1×15?

A. $10 \times 0.1 \times 15$ B. $(10 + 1) \times 15$ C. $(10 + 0.1) \times 15$ D. $10 + 0.1 \times 15$

Exploration 1

Work out each of the following.

(1) 12.3×9.9

(2) 5.14×1.01

Exploration 2

1 What is the value of $4.5 \times 5.5 + 4.5 \times 4.5$?

A. 36.5 B. 45 C. 50 D. 90 E. 100

2 Work out each of the following.

(1) $0.13 \times 78 + 0.13 \times 21 + 0.13$

(2) $6.66 \times 17 + 6.66 \times 82 + 6.66$

Exploration 3

1 What is the value of $2.017 \times 2016 - 10.16 \times 201.7$?

A. 2.016 B. 2.017 C. 20.16 D. 2016 E. 2017

2 Calculate $999.9 \times 0.28 - 0.6666 \times 370$

Learn and Discover

1 Calculate $\left(\frac{7}{9} - \frac{5}{6} + \frac{5}{18}\right) \times (-18) =$ ______

2 Calculate $\frac{4}{5} \times \left(-\frac{5}{13}\right) - \frac{3}{5} \times \left(-\frac{5}{13}\right) - \frac{5}{13} \times \left(-1\frac{3}{5}\right) =$ ______

Exploration 4

1 Work out each of the following.

(1) $\left(-\frac{3}{4} + 1\frac{1}{3} - 2.5\right) \times (-24)$

(2) $\left(-\frac{3}{4} - \frac{5}{9} + \frac{7}{12}\right) \div \frac{1}{36}$

2 Work out each of the following.

(1) $-7 \times \left(-\frac{4}{19}\right) + 13 \times \left(-\frac{4}{19}\right) - 6 \times \left(-\frac{4}{19}\right)$

(2) $(-0.25) \times \left(-5\frac{1}{2}\right) + \frac{1}{4} \times (-3.5) + \left(-\frac{1}{4}\right) \times (-2)$

Practice

Calculate $\frac{1}{7} \times \left(-\frac{6}{5}\right) + 1.2 \times \left(-\frac{2}{7}\right) + \frac{4}{7} \times \left(-1\frac{1}{5}\right)$

Exploration 5

Work out each of the following.

(1) $\frac{1}{5} \times \left(-75\frac{5}{6}\right)$

(2) $-99\frac{98}{99} \times 99$

Challenge

1 Calculate $7 \times 11 \times 10 \times \left(\frac{1}{7} + \frac{1}{11} + \frac{1}{10}\right)$

2 Calculate $84 \times \frac{13}{31} + 42 \times \frac{22}{31} + 28 \times \frac{21}{31}$

After Class

Basic Exercises

1. What is the value of $(12340 + 12.34) \div 1234$?

A. 100.01　　B. 100.1　　C. 10.001　　D. 10.01　　E. 10.1

2. Calculate $7.8 \times 3.3 + 7.8 \times 7.7 - 7.8 =$ ______

3. Work out each of the following.

(1) $\frac{4}{3} \times \left(\frac{5}{8} - \frac{1}{4}\right)$

(2) $\left(\frac{5}{12} - \frac{1}{5}\right) \times -60$

4 Work out each of the following.

(1) $\left(\frac{5}{12}+\frac{2}{3}-\frac{3}{4}\right)\times(-24)$

(2) $\left(1\frac{3}{4}-\frac{7}{8}-\frac{7}{12}\right)\div\left(-\frac{7}{8}\right)$

5 Work out each of the following.

(1) $-14\frac{5}{8}\times\frac{5}{9}+\frac{5}{9}\times-3\frac{3}{8}$

(2) $-\frac{3}{4}\times-98+-\frac{3}{4}\times-2$

6 Calculate $-4\times\left(-\frac{8}{9}\right)+(-8)\times\left(-\frac{8}{9}\right)+12\div\left(-\frac{9}{8}\right)=$ ______

7 Calculate $-\frac{1}{4}\times(-19)-\frac{1}{2}\times 19-\frac{3}{4}\times(-19)$

8 Calculate $(-5) \times \left(+7\frac{1}{3}\right) + 7 \times \left(-7\frac{1}{3}\right) - (+12) \times \left(-7\frac{1}{3}\right) =$ ______

9 Calculate $6.31 \times 1.1 + 3.3 \times 1.23 =$ ______

10 Calculate $20.07 \times 39 + 200.7 \times 4.1 + 40 \times 10.035 =$ ______

Extensive Exercises

1 Calculate $2 \times \frac{4}{5} + 3 \times \frac{4}{5} + 11 \times \frac{2}{9} - 2 \times \frac{2}{9} =$ ______

2 Calculate $13 \times \left(-\frac{11}{5}\right) + 7 \times \left(-\frac{11}{5}\right) + 5 \times \left(-\frac{16}{7}\right) + 2 \times \left(-\frac{16}{7}\right) =$ ______

3 Calculate $39 \times \frac{144}{147} + 144 \times \frac{84}{147} + 48 \times \frac{72}{147} =$ ______

Extensive Challenge

1 Calculate $\left(-9\frac{24}{25}\right) \times (-125) =$ ______

2 Work out each of the following.

(1) $57 \times \frac{55}{56} + 27 \times \frac{27}{28}$

(2) $2019 \times \frac{2017}{2018} - 1010 \times \frac{1008}{1009}$

Lesson 6

Prime Factors/LCM/HCF

Target Grade

Knowledge Points	Target Grade	Topic Area
Factors, multiples and prime numbers	1	Number
Prime factors	4	
LCM/HCF	IMC: difficulty 4	

Objectives

- Be able to use the concepts and vocabulary of prime numbers, factors, multiples and prime factorisation, including using product notation and the unique factorisation property.
- Be able to find the lowest common multiple or highest common factors of two or more numbers.

Let's Get Ready

1. Which of the following numbers is not a prime number?

A. 97　　B. 43　　C. 87　　D. 59　　E. 83

2. Which of the following are multiples of 6?

A. 3　　B. 42　　C. 28　　D. 6　　E. 80

3. Write down all the factors of 200

In Class

Factors, Multiples and Prime Numbers

Notes

1. Factors and Multiples

(1) A **multiple** of a number is just a number that appears in its times table.

(2) A **factor** of a number is an ________ that divides exactly into another number without ________________. A pair of **factors** can be multiplied together to give the original number.

e.g. If we want to list the factors of the number 12, we need to find the numbers that 12 can be divided by. We can write them out as pairs:

$1 \times 12 = 12$, $2 \times 6 = 12$, $3 \times 4 = 12$

We can say that 1, 2, 3, 4, 6 and 12 are all **factors** of 12

Another way of saying this is that 12 is a **multiple** of each of its **factors**.

2. Understanding Prime Numbers

(1) A **prime number** is a special type of number that can only be divided by __________; they only appear in their own times table.

(2) Another way of describing **prime numbers** is that they are numbers with only ________________.

(3) There are **some important facts** you should know about **prime numbers**:

① The number 1 is **not** a prime number – it has only **one positive factor** (itself).

② The number 2 is the **only even** prime number.

③ Except for 2 and 5, all prime numbers **end in** 1, 3, 7 or 9

④ Not all numbers ending in 1, 3, 7 and 9 are prime.

⑤ There is no pattern in the series of prime numbers.

The **prime numbers** between 1 and 100 are displayed below:

Find Prime Numbers

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

3. Finding Prime Numbers

To show whether a number is prime or not, there are a couple of things to bear in mind that make this process a lot easier:

(1) Ignore any numbers that end in $0, 2, 4, 6$, and 8 – they are all divisible by 2

(2) Ignore any numbers that end in 5– they are all divisible by 5

(3) If a number does not end in $0, 2, 4, 5, 6$, and 8, try to divide it by other prime factors $3, 7, 11, 13 \ldots$ one by one until the quotient is less than the divisor.

Exploration 1

Nidah writes down two different prime numbers.
She adds together her two numbers.
Her answer is a square number less than 30
Find two prime numbers that Nidah could have written down.

Notes

4. Prime Factorisation

Prime factors are the 'building blocks' of other larger numbers. We can divide a number into its **factors** and keep dividing these **factors** by their **factors** until we are left with nothing but **prime numbers** – this is called ________________.

Learn and Discover

Find the prime factorisation of 300 using the short division method.

Exploration 2

Find prime factorisations for the following numbers.

a. 28

b. 75

c. 10000

d. 378

HCF/LCM

Notes

1. Highest Common Factor (HCF)

The **Highest Common Factor (HCF)** of two numbers is the ________ number which is a factor of both of them.

Consider the factors of 6 and 18

Factors of 6 are ________ and factors of 18 are ____________. Factors in common are________. We see that the lowest common factor of 6 and 18 is 1 and the highest common factor is 6. The lowest common factor is not useful as it is always 1 for any two or more numbers. So we only care about the highest common factor.

2. Lowest Common Multiple (LCM)

The **Lowest Common Multiple (LCM)** of two numbers is the ________ number which is a multiple of these numbers.

Consider the multiples of 6 and 18

Multiples of 6 are $6, 12, 18, 24, 36...$ and multiples of 18 are $18, 36, 54....$ Multiples in common are $18, 36....$ We see that the lowest common multiple of 6 and 18 is 18

Warm Up

$X = 2^2 \times 3^2 \times 7$

$Y = 2 \times 3^2 \times 5^2$

(1) Find the highest common factor of X and Y.

(2) Find the lowest common multiple of X and Y.

Learn and Discover

1 We can use short division method to express prime factorisation in a simpler way, as shown in the example below.

$$90 = 2 \times 3 \times 3 \times 5$$
$$120 = 2 \times 2 \times 2 \times 3 \times 5$$

$$\begin{array}{r|rr} 2 & 90 & 120 \\ \hline 3 & 45 & 60 \\ \hline 5 & 15 & 20 \\ \hline & 3 & 4 \end{array}$$

HCF: $2 \times 3 \times 5 = 30$ HCF: $2 \times 3 \times 5 = 30$

Use short division method to find the highest common factor of 75 and 105

2 We can use short division method to express prime factorisation in a simpler way, as shown in the example below.

$$48 = 2 \times 2 \times 2 \times 2 \times 3$$
$$60 = \cancel{2} \times \cancel{2} \times \cancel{3} \times 5$$

$$\begin{array}{r|rr} 2 & 48 & 60 \\ \hline 2 & 24 & 30 \\ \hline 3 & 12 & 15 \\ \hline & 4 & 5 \end{array}$$

LCM: $2 \times 2 \times 2 \times 2 \times 3 \times 5 = 240$ LCM: $2 \times 2 \times 3 \times 4 \times 5 = 240$

Use short divison method to find the lowest common multiple of 42 and 54

Notes

3. Use Prime Factorisation to Find the HCF/LCM of Two Numbers

Step 1: Write the prime factorisation of these two numbers using short division method or in expanded form.

Step 2: Multiply all their common factors together to obtain the highest common factor.

Step 3: Multiply all the factors together, and all the common factors only need to be multiplied ______ to obtain the lowest common multiple.

Exploration 3

Find the HCF and LCM of each pair of numbers.

	26 and 39	45 and 110	14 and 28	5 and 9
HCF				
LCM				

Exploration 4

1. There are 72 pears and 54 oranges, which are shared among several children. Each child will get the same number of pears and the same number of oranges.
How many children are there at most?
How many pears does each child get in that case?

2 Alarm A rings every 6 min. Alarm B rings every 8 min. They rang together at $7:30$ a.m. At what times will both alarms ring together before $8:30$ a.m.?

Challenge

Martin thinks of two numbers.
He says,
"The Highest Common Factor (HCF) of my two numbers is 6
The Lowest Common Multiple (LCM) of my two numbers is a multiple of 15"
Write down **three** possible groups of numbers that Martin is thinking of.

Learn and Discover

1 $A = 2^2 \times 3^2$
$B = 2^4 \times 3$
$C = 2^2 \times 3 \times 5$

(1) Write the prime factorisation of A, B and C in expanded form.

(2) Circle all the common prime factors of A, B and C.

(3) Find the highest common factor of A, B and C.

2 $A = 3 \times 5 \times 7$

$B = 2 \times 3^2$

$C = 2^2 \times 5$

(1) Find the lowest common multiple of A and B.

(2) Find the lowest common multiple of A, B and C.

Notes

4. Find the HCF/LCM of three numbers

Step 1: Write the prime factorisation of these three numbers in expanded form.
Step 2: Circle all the common factors which appear in all the lists.
Step 3: Multiply all their common factors together to obtain the **highest common factor**.
Step 4: To get the LCM of three numbers, we can find the LCM of any two of them first, and then find the LCM of the result and the remaining third number.

Exploration 5

1 $D = 2 \times 3^2 \times 5^2$

$E = 2^2 \times 3 \times 5$

$F = 2 \times 5 \times 11$

(1) Find the highest common factor of D, E and F.

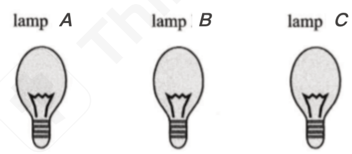
(2) Find the lowest common multiple of D, E and F.

2 Find the LCM and HCF of 462, 147 and 56

Exploration 6

1 Marta has collected 42 apples, 60 apricots and 90 cherries. She wants to divide them into identical piles using all of the fruit and then give a pile to some of her friends. What is the largest number of piles she can make?

2 Here are three lamps.



Lamp A flashes every 36 seconds.
Lamp B flashes every 45 seconds.
Lamp C flashes every 120 seconds.
The three lamps start flashing at the same time.
How many times in one hour will the three lamps flash at the same time?

After Class

Basic Exercises

1 (1) Write 420 as a product of its prime factors.

Answer: ______

(2) Write 1620 as a product of its prime factors.

Answer: ______

2 $P = 2^4 \times 5 \times 7$

$Q = 2^2 \times 5^2 \times 7$

(1) Find the highest common factor of P and Q.

Answer: ______

(2) Find the lowest common multiple of P and Q.

Answer: ______

3 $X = 2^2 \times 5^2$

$Y = 2 \times 3 \times 5^3$

(1) Find the highest common factor of X and Y.

Answer: ______

(2) Find the lowest common multiple of X and Y.

Answer: ______

4 Find the HCF and LCM of each pair of numbers.

	70 and 84	52 and 78	17 and 34	7 and 8
HCF				
LCM				

5 There are 60 sandwiches and 75 burgers, which are shared among several people. Each person will get the same number of sandwiches and the same number of burgers.
How many people are there at most?
How many burgers does each person get in that case?

6 Two lighthouses flash their lights every 36 seconds and 30 seconds. Given that they flash together at 08 : 00 am, at what time would the two lighthouses flash their lights together again?

7 Find the highest common factor and lowest common multiple of the numbers below.

$A = 2^4 \times 3$

$B = 2^3 \times 3 \times 5^2$

$C = 2^2 \times 3$

Answer: HCF = ______

LCM = ______

8 Find the HCF and LCM of 32, 152 and 600

Answer: HCF = ______

LCM = ______

9 Eddie has three wooden sticks with lengths of $54\ \text{cm}$, $63\ \text{cm}$ and $144\ \text{cm}$, and he wants to cut them into pieces of equal length without leaving any surplus. What is the longest possible length of each piece?

Answer: ______ cm

10 Buses to Ayton leave the station every 12 minutes.
Buses to Bleeford leave the station every 20 minutes.
Buses to Canterbury leave the station every 8 minutes.
Buses to all three places leave at $07:05$ am.
How many times between 7 am and 12 pm will the buses to three destinations leave the station together at the same time?

Answer: ______ times

Extensive Exercises

1 Ali is planning a party. He wants to buy some cakes and some sausage rolls.
The cakes are sold in boxes. There are 12 cakes in each box. Each box of cakes costs £2.50. The sausage rolls are sold in packs. There are 8 sausage rolls in each pack. Each pack of sausage rolls costs £1.20.
Ali wants to buy more than 60 cakes and more than 60 sausage rolls. He wants to buy exactly the same number of cakes as sausage rolls. What is the least amount of money Ali will have to pay?

2 $A = 3^5 \times 5 \times 7^3$
$B = 2^3 \times 3 \times 7^4$
$C = 2^p \times 5^q \times 7^r$
Given that
the HCF of B and C is $2^3 \times 7$
the LCM of A and C is $2^4 \times 3^5 \times 5^2 \times 7^3$
find the value of p, the value of q and the value of r.

Answer: $p =$ ______
$q =$ ______
$r =$ ______

3 William thinks of two numbers.
He says,
"The Highest Common Factor (HCF) of my two numbers is 12
The Lowest Common Multiple (LCM) of my two numbers is a multiple of 9"
Write down **three** possible groups of numbers that William is thinking of.

Extensive Challenge

1. Grandma bakes many cookies for her grandchildren who are going to visit her in the afternoon. She has forgotten whether $2, 3, 5$ or all 6 of her grandchildren will visit. She wants all the cookies eaten and each grandchild to get the same amount of cookies. To be prepared for all the possibilities, what is the smallest number of cookies she should bake?

2. A bus timetable shows the following information.
   - A bus following route A leaves for the station every 6 minutes.
   - A bus following route B leaves for the station every 8 minutes.
   - A bus following route C leaves for the station every 12 minutes.
   - A bus following route D leaves for the station every 15 minutes.
   - Four buses leave together at $08:30$ am.

   When is the next time one of each bus is timetabled to leave at the same time?

Lesson 7

Percentages

Target Grade

Knowledge Points	Target Grade	Topic Area
Percentages of amounts	3	Number
Percentage change	5 JMO: difficulty 4	

Objectives

- Be able to define percentage as number of parts per hundred.
- Be able to interpret percentages and percentage changes as a fraction or a decimal and interpret these multiplicatively.
- Be able to express one quantity as a percentage of another and compare two quantities using percentages.
- Be able to work with percentages greater than 100%.
- Be able to solve problems involving percentage change, including percentage increase/decrease and original value problems.

Let's Get Ready

1. Write 0.6 as a percentage.

Answer: ______ $\%$

2. Write $\frac{4}{50}$ as a percentage.

Answer: ______ $\%$

3. Write 40% as a fraction in its simplest form.

Answer: ______

In Class

Percentage of Amounts

Notes

1. Percentages

Percentage means 'number of parts per hundred' and is denoted by the ________.

For example, 50% of a number means 50 parts out of 100

Since 50 is half of 100, 50% also means ______________________.

Percentage could be _________ or __________.

Percentage also could exceed _________.

Learn and Discover

Sandra scored 30 marks in a test with a total of 150 marks.

What percentage did she score in this test?

Notes

2. Expressing one quantity as a percentage of total

To express one quantity as a percentage of the total, use the following equation:

$$\text{Percentage} = \frac{\text{Value}}{\text{Total}} \times 100$$

Learn and Discover

200 people took course A and 125 people took course B.
188 people passed course A and 110 people passed course B.
Which course has a higher pass rate?

Notes

3. Comparing the percentage of two quantities

To compare two different quantities, express each as a percentage of its total first, then compare the percentages.

Exploration 1

Karen scored 32 out of 80 in a maths test.
She scored 38% in an English test.
Did Karen get a higher percentage in maths or in English?

Learn and Discover

A chair normally costs £300
In a sale, the price is reduced to 70% of the normal price.
What is the sale price?

Notes

4. Percentage of an amount
To calculate the **percentage of an amount**, we convert the percentage to a decimal or fraction and then multiply this by the amount.

Exploration 2

1 Calculate 20% of 30% of 40% of £50

2 Find the sum of 50% of 2006 and 2006% of 50

A. 1013.3 B. 1053 C. 1103.3 D. 1504.5 E. 2006

Challenge

At a certain school, 48% of the pupils are girls.
25% of the girls and 50% of the boys at this school travel to school by bus.
What percentage of the pupils in the school travel by bus?

A. 37% B. 38% C. 62% D. 73% E. 75%

Percentage Changes

Learn and Discover

The normal price of a television is £350
In a sale, this price is reduced by 30%
Work out the sale price of the television.

Notes

1. Percentage increase and decrease

For a **percentage increase**, the multiplier (decimal or fraction) will be ____________ 1

For a **percentage decrease**, the multiplier (decimal or fraction) will be _________ 1

A **decimal multiplier** is a decimal number used to calculate the percentage of an amount, or to increase or decrease an amount by a percentage. It can be calculated as follows:

- A **percentage increase** of 15% is $(100+15)\%$ of the original value.

Decimal multiplier $= \dfrac{115}{100} = 1.15$

- A **percentage decrease** of 15% is $(100-15)\%$ of the original value.

Decimal multiplier $= \dfrac{85}{100} = 0.85$

Learn and Discover

Felicity buys a dress in a sale.
It is reduced by 30% down to a price of £42
Work out the original price of the dress.

Notes

2. Reverse percentages

Reverse percentages are where we are given a value or an amount that has increased or decreased by a certain percentage, and then we have to use this information to calculate the **original amount**.

Basic steps to find the original value:

Step 1: Find the decimal multiplier of the original value (e.g. the decimal multiplier for an increase of 10% is 1.1, and for a decrease of 10% is 0.9)

Step 2: Divide the given amount by this decimal multiplier to find the original value.

Exploration 3

The price of a train ticket increased by 25% and then decreased by 10% in a special offer.
It was then priced at £45
What was the original price of the ticket?

Learn and Discover

Calculate the percentage decrease when a car goes down in value from £8000 to £2000

Notes

3. Percentage change

Percentage change is used to find the change in a value as a percentage. It can be a ____________________ or a ____________________. It helps us to understand how much something has changed from what we started with.

$$\text{Percentage Change} = \frac{\text{Change}}{\text{Original}} \times 100$$

Note: To determine if the percentage change is a percentage decrease or a percentage increase:

If the original value is higher than the final value, then it is a percentage decrease.

If the original value is lower than the final value, then it is a percentage increase.

Exploration 4

Lauren's weight decreases from 60 kg to 48 kg.

Calculate the percentage decrease in Lauren's weight.

Challenge

Jack and Sadia work for a company that sells boxes of breakfast cereal.
The company wants to have a special offer.
Here is Jack's idea for the special offer.

Put 25% more cereal into each box and keep the price the same.

Here is Sadia's idea.

Reduce the price and keep the amount of cereal in each box the same.

Sadia wants her idea to give the same value for money as Jack's idea.
By what percentage does she need to reduce the price?

Answer: ______ %

Learn and Discover

Emily buys a pack of 12 bottles of water for £5
She sells all 12 bottles for 50p each.
Work out Emily's percentage profit.

Notes

4. Percentage Profit

Profit = Total selling price - Total cost

Percentage profit is expressing a profit as a ____________. This makes it easier to _________________________. To do this you need to find the profit and change it as a percentage of the original cost price.

$$\text{Percentage Profit} = \frac{\text{Profit}}{\text{Total Cost}} \times 100$$

Exploration 5

Renee buys $5\ \mathrm{kg}$ of sweets for £10
She packs all the sweets into bags, putting $250\ \mathrm{g}$ of sweets into each bag.
She sells each bag of sweets for $65\mathrm{p}$ and sells all the bags.
Work out her percentage profit.

Exploration 6

Helen has two stamps for sale: one is a rare stamp (a Penny Black) and the other is a more common stamp (a Penny Red).

She sells them for £300 each.

Based on their original cost prices, she made a profit of 25% on the Penny Black, and a loss of 25% on the Penny Red.

(1) Calculate the profit she made on the Penny Black.

(2) Calculate the loss she made on the Penny Red.

Challenge

In 2003, Jerry bought a house.

In 2007, he sold the house to Mia at a profit of 20%

In 2012, Mia sold the house for £162000, making a loss of 10%

Work out how much Jerry paid for the house in 2003

After Class

Basic Exercises

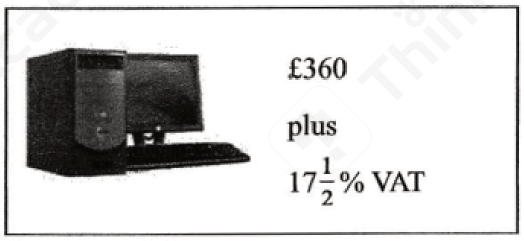
1 Which of the following has the greatest value?

A. 50% of 10 B. 40% of 20 C. 30% of 30 D. 20% of 40 E. 10% of 50

2 A computer costs £ 360 plus $17\frac{1}{2}\%$ VAT.

Calculate the total cost of the computer.

Answer: £ ______



3 John needs 4 tyres for his car.

He pays for 3 tyres and gets one tyre free.

The tyres cost £65 each plus VAT at 20%

Work out how much in total John pays for the tyres.

Answer: £ ______

4 A cereal company has decreased the amount of cereal in a box from $1.2 \mathrm{~kg}$ to $900 \mathrm{~g}$.
Work out the percentage decrease in the amount of cereal per box.

Answer: ______ $\%$ decrease

5 When a loaf of bread is baked, it loses 12% of its weight.
Claire wants the baked loaf of bread to weigh $1.1 \mathrm{~kg}$.
Work out the weight of the loaf of bread before it is baked.

6 In a sale, normal prices are reduced by 20%
A washing machine has a sale price of £464
Calculate the reduction in the price of the washing machine.

Answer: £ ______

7 Sarah is a ski jumper.
The maximum length of skis she can use is $4.25 \mathrm{~m}$, which is 150% more than her height.
Work out Sarah's height.

8 Peter bought a table for £800 and sold it for £1000
Calculate his percentage profit.

Answer: ______ $\%$

9 The selling price of a smartphone is £420
Given that the percentage profit is 40%, calculate the cost price of the smartphone.

Answer: £ ______

10 Sean buys 24 chocolate bars for £10
He sells them for 50p each.
Work out his percentage profit.

Answer: ______ $\%$

Extensive Exercises

1 A trader buys six watches at £25 each.
He scratches one of them, so he sells that one for £10
He sells the other five for £40 each.
Find his percentage profit.

Answer: ______ %

2 A property owner has two houses for sale – one in a fashionable area (called Costaplenty), and one in a less fashionable area (called Thisledo).
He manages to sell them for £156000 each.
Based on what he originally paid for them, Costaplenty made him a profit of 20%, while Thisledo made him a loss of 20%

(1) Calculate the profit made on Costaplenty.

(2) Calculate the loss made on Thisledo.

3 On Monday, the Pied Piper caught 1000 rats in a city.
On Tuesday, he caught 10% fewer than on Monday.
On Wednesday, he caught 20% more than on Tuesday.
On Thursday, he caught 30% fewer than on Wednesday.
On Friday he rested.
How many rats did he catch in total that week?

Answer: ______

Extensive Challenge

1. The price of a train ticket increased by 5% and then decreased by 20% in a special offer. It was then £4 less expensive than its original price.
What was the original price of the ticket?
A. £8.60 B. £13 C. £20.40 D. £25 E. £26.40

2. In a sale, the price of a computer is reduced by 20%
At this reduced price, the shopkeeper still makes a profit of 20%
What would have been his percentage profit if the computer had been sold at full price?

Answer: ______ %

Lesson 8

Indices

Target Grade

Knowledge Points	Target Grade	Topic Area
Simplifying algebraic expressions by multiplying indices	4 IMC: difficulty 3	Algebra-Manipulation
Simplifying algebraic expressions by dividing indices		

Objectives

- Be able to use, interpret, simplify and manipulate index notation.
- Be able to use laws of indices for multiplying powers, dividing powers and raising a power to a power.
- Be able to deal with a power of zero.
- Be able to express a power using a different base.

Let's Get Ready

1. Work out the value of 3^5

Answer: ______

2. Given that $2^m = 64$, work out the value of m.

A. 6　　B. 7　　C. 8　　D. 9

3. Which of the following has the least value?

A. $2^1 - 1^2$　　B. $3^2 - 2^3$　　C. $4^3 - 3^4$　　D. $5^4 - 4^5$

In Class

Simplify Algebraic Expressions by Multiplying Indices

Learn and Discover

Consider the following:

$5^2 = 5 \times 5$

$5^3 = 5 \times 5 \times 5$

Calculate $5^2 \times 5^3$ and write your answer in index form.

$5^2 \times 5^3 =$ ______

What did you find out?

Notes

1. Multiplication law

The **multiplication law** states that when you multiply powers with the same base, you ______________.

$$a^b \times a^c = a^{b+c}$$

e.g. $6^3 \times 6^4 = 6^7$

Exploration 1

Simplify the following and give your answers in index form.

(1) $4^3 \times 4$

(2) $(-3)^9 \times (-3)^{91}$

(3) $x^2 \times x^3$

(4) $3m^2 r \times 4m^3 r^6$

(5) $4m^3 q^2 p^5 \times 10m^2 q^2 p$

Challenge

Given that $4^x + 4^x + 4^x + 4^x = 4^{16}$, what is the value of x?

A. 2 B. 4 C. 8 D. 12 E. 15

Learn and Discover

Recall that $x^3 = x \times x \times x$.

Consider $(2^4)^3$

How can we simplify this?

First, by the definition of a cube, $(2^4)^3$ can be rewriten as ______.

Then, by the multiplication law of indices, it can be simplified as ______.

Notes

2. Powers of a power law

The **power of a power law** states that when you raise one power to another (i.e. the power of a power), the indices are__________:

$$(a^b)^c = a^{bc}$$

e.g. $(7^9)^8 = 7^{9\times 8} = 7^{72}$

Exploration 2

Answer the following questions.

(1) Simplify $(x^2)^5$

(2) Given that $3^n = 2$, find the value of $(3^4)^n$

Challenge

If $x = 2^p$ and $y = 2^q$, express the following in terms of x and/or y:

(1) 2^{p+4}

(2) 2^{2q}

(3) 2^{3p+2q}

Learn and Discover

Consider the expressions $(2 \times 4)^3$ and $2^3 \times 4^3$

Calculate the value of each expression.

What do you notice?

Notes

3. Product law

In general, if there is a complex term inside brackets, apply the power to each factor individually:

$$(ab)^n = a^n b^n$$

e.g. $(15)^2 = 3^2 \times 5^2$

Exploration 3

Simplify each of the following.

(1) $(p^2r^5)^2$

(2) $(3w^2y)^3$

Learn and Discover

Write $9^2 \times 27^3$ as a power of 3

First, $9 = 3^2$, so $9^2 =$ ______

Similarly, $27 = 3^3$, so $27^3 =$ ______

Therefore, $9^2 \times 27^3 =$ ______

Notes

4. Changing the base

To multiply powers with different bases, instead of evaluating each term, we can express them with a **common base**. We can then apply the laws of indices (such as the **power of a power law** and the **multiplication law**) to simplify.

The table below shows some common powers that are useful to memorise:

Base\Power	2	3	4	5	6
2	4	8	16	32	64
3	9	27	81	243	×
4	16	64	256	×	×
5	25	125	625	×	×

Exploration 4

Answer the following questions.

(1) Write $\left(16^3\right)^4$ as a single power of 2

(2) Given that $2^p \times 8^q = 2^n$, express n in terms of p and q.

Challenge

Given that $8^m = 27$, work out the value of 4^m

A. 3

B. 4

C. 9

D. 13.5

E. There is no such m

Simplify Algebraic Expressions by Dividing Indices

Learn and Discover

Consider how to simplify $2^7 \div 2^3$

First, write the expression as a fraction: _____

Then, write the expanded form of the terms: _____

This gives _____ which is _____ in index form.

Notes

1. Division law

The division law states that when you **divide terms with the same base**, you ____________________:

$$a^b \div a^c = a^{b-c}\,(a \neq 0)$$

e.g. $\frac{19^8}{19^7} = 19^{8-7} = 19^1 = 19$

2. Power of zero

If the index is 0, the value is 1 (e.g. $9^0 = 1$).

$$\boldsymbol{a^0 = 1(a \neq 0)}$$

Consider using the division law to prove this:

$$a^0 = a^{m-m} = \_\_\_\_\_\_\_\_\_\_ = \_\_\_\_\_\_ = \_\_\_.$$

What about 0^0 ? Since division by zero is undefined, 0 is mathematically *"indeterminate"*.

3. Summary

Laws	Explanations
$a^1 = a$	
$a^b \times a^c = a^{b+c}$	$a^3 \times a^2 = (a \times a \times a) \times (a \times a) = a^{3+2} = a^5$
$(ab)^n = a^n b^n$	$(ab)^3 = a \times b \times a \times b \times a \times b = (a \times a \times a) \times (b \times b \times b) = a^3 b^3$
$(a^b)^c = a^{bc}$	$(a^3)^2 = (a \times a \times a) \times (a \times a \times a) = a^{3\times 2} = a^6$
$a^b \div a^c = a^{b-c}$	$a^5 \div a^2 = \frac{a \times a \times a \times a \times a}{a \times a} = a^{5-2} = a^3$
$a^0 = 1(a \neq 0)$	$a^0 = a^{2-2} = a^2 \div a^2 = 1$

Exploration 5

Simplify the following expressions.

(1) $n^4 \div n^2$

(2) $\dfrac{x^{17}}{x^5}$

(3) $4n^2p^3 \div 2n^2p$

(4) $\dfrac{r^3q^7m^4}{5r^2q^5}$

Exploration 6

1 Which of the following is equal to $\dfrac{3^9}{9^3}$?

A. 3 B. 9 C. 27 D. 81 E. 243

2 Work out half of 4^{2022}

A. 4^{1011} B. 2^{4044} C. 4^{2021} D. 2^{4043} E. 2^{1011}

Challenge

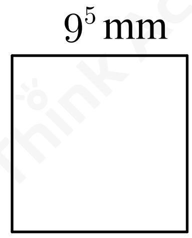
In the calculation shown, how many times does the term 2018^2 appear inside the square root to make the calculation correct?

$$\sqrt{2018^2+2018^2+\cdots+2018^2}=2018^{10}$$

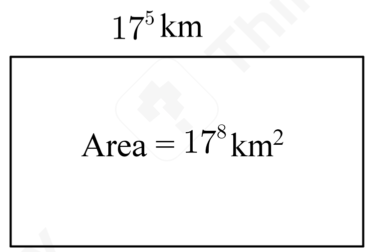
After Class

Basic Exercises

1 A square has a side length of 9^5 mm.
Calculate the area of the square.
Give your answer as a single power of 9



2 A rectangle has an area of 17^8 km^2 and a length of 17^5 km.
Calculate the width of the rectangle.
Give your answer as a single power of 17



3 (1) Simplify $p^2 \times p^5$

(2) Simplify $g^6 \div g^4$

(3) Simplify $(k^3)^2$

4 Simplify the following expressions.

(a) $2x^2y \times 3xy^4$

(b) $(3m^7)^2$

(c) $(3x^2y^4)^3$

5 Simplify $\dfrac{6a^5b^3}{3a^2b^2}$

6 For each equation, find the value of m.

(1) $13^m \times 13^7 = 13^{20}$

(2) $(-2)^{-5} \times (-2)^m = (-2)^{10}$

(3) $4^m \times 4^m \times 4^m = 4^{15}$

7 Simplify $(x^2 \times x^3)^2$

8 $2^y = 4^4$

Write down the value of y.

Answer: $y =$ ______

9 Write $2^{15} \times 8^2 \times 16^3$ as a single power of 2

10 Here are five number cards.

16	4^3	32	2^2	$(2^3)^2$

(1) Which two cards are equal in value?

(2) Which two cards have a product equal to 2^9?

Extensive Exercises

1 Find the value of $\frac{10^5}{5^5}$

Answer: ______

2 Work out one-ninth of 3^{99}

Give your answer in index form.

3 Given that $a^b = \frac{1}{8}$, find the value of a^{3b}.

Extensive Challenge

1 Which of the following shows the correct order of the three numbers 81^{31}, 27^{41} and 9^{61}?

A. $81^{31} > 9^{61} > 27^{41}$
B. $27^{41} > 81^{31} > 9^{61}$
C. $9^{61} > 27^{41} > 81^{31}$
D. $27^{41} > 9^{61} > 81^{31}$
E. $81^{31} > 27^{41} > 9^{61}$

2 Given that $5^{p} = 9$, $9^{q} = 12$, $12^{r} = 16$, $16^{s} = 20$ and $20^{t} = 25$, find the value of $pqrst$.

Answer: $pqrst =$ ______

Stage Test (L5-L8)

1 What is the value of $(74100 + 7.41) \div 741$?

A. 10.01 B. 100.1 C. 100.01 D. 1000.01 E. 10.1

2 Calculate $5.9 \times 4.2 + 5.9 \times 6.8 - 5.9$

3 Calculate each of the following.

(1) $\frac{5}{2} \times \left(\frac{7}{10} - \frac{1}{5}\right)$

(2) $\left(\frac{7}{15} - \frac{1}{4}\right) \times (-60)$

4 (1) Write 630 as a product of its prime factors.

(2) Write 2250 as a product of its prime factors.

5 $X = 2^5 \times 3 \times 11$

$Y = 2^3 \times 3^2 \times 11$

(1) Find the highest common factor (HCF) of X and Y.

(2) Find the lowest common multiple (LCM) of X and Y.

Give your answers in index form.

6 There are 54 chocolate bars and 90 packets of crisps, which are shared among several students.

Each student will get the same number of chocolate bars and the same number of packets of crisps.

(1) What is the greatest number of students there can be?

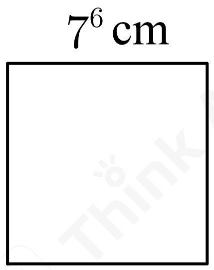
(2) How many packets of crisps does each student get in that case?

7 Which of the following has the greatest value?

A. 70% of 10 B. 60% of 20 C. 50% of 30 D. 40% of 40 E. 30% of 50

8 A supermarket has decreased the amount of washing powder in a box from $1.5\ \text{kg}$ to $1200\ \text{g}$.
What is the percentage decrease in the amount of washing powder per box?

9 A square has a side length of 7^6 cm.
Calculate the area of the square.
Give your answer as a power of 7



10 Simplify each of the following.

(1) $4a^3b \times 3ab^5$

(2) $(4n^6)^2$

(3) $(2x^4y^3)^3$

Exercises' Answers

Lesson 1 Rational Numbers and Number Line

Basic Exercises

1. B
2. 37
3. -2
4. -3
5. 14
6. C
7. A
8. C
9. 6; 0
10. (1) $<$
    (2) $<$
    (3) $>$

Extensive Exercises

1. (1) -5
   (2) 3; 7
2. B
3. 10

Extensive Challenge

1. A
2. (1) 4
   (2) ① 4 or -2
   ② 3 or -7
   ③ 2 or -6

Lesson 2 Addition and Subtraction of Rational Numbers

Basic Exercises

1. (1) -110
   (2) 4
   (3) -45
   (4) 0
   (5) 25
2. (1) -37.3
   (2) 0.6
   (3) -27.6
3. (1) 68
   (2) -73
   (3) 135
   (4) -18
4. (1) 20.1
   (2) -25.7
   (3) -33.2

5. (1) $-3\frac{4}{5}$
   (2) $-5\frac{7}{9}$
   (3) $-1\frac{1}{12}$
   (4) $9\frac{3}{16}$
6. (1) $-\frac{4}{35}$
   (2) $7\frac{1}{12}$
   (3) $3\frac{1}{15}$
   (4) $2\frac{17}{24}$
7. (a) 19.7
   (b) -7
   (c) 8
   (d) -7.3
8. (1) 0
   (2) 3
   (3) -3
9. (1) South, 13 km
   (2) 3.35 litres
10. -36

Extensive Exercises

1. C
2. A
3. (1) -13
   (2) 3

Extensive Challenge

1. -40
2. A

Lesson 3 Multiplication and Division of Rational Numbers

Basic Exercises

1. (1) -56
   (2) 7
   (3) 150
2. (1) $-\frac{3}{4}$
   (2) $\frac{11}{6}$
   (3) 3
   (4) $-\frac{11}{6}$
3. (1) $-\frac{16}{3}$; (2) $\frac{2}{3}$; (3) $-\frac{3}{14}$; (4) -54; (5) -8; (6) $\frac{2}{9}$

4. (1) $-\frac{4}{3}$
   (2) $\frac{1}{4}$
   (3) $-\frac{16}{49}$
   (4) $\frac{5}{3}$
   (5) $-\frac{3}{2}$
   (6) 20
5. (1) 144
   (2) -120
   (3) 81
   (4) -70
6. $-\frac{1}{6}$
7. (1) $\frac{4}{27}$
   (2) -2
   (3) $\frac{7}{9}$
8. (1) $\frac{9}{4}$
   (2) 49
   (3) 270
9. -20
10. -1

Extensive Exercises

1. E
2. -2；12
3. -3；-2；4

Extensive Challenge

1. 0
2. 256

Lesson 4 Powers and Numbers in Standard Form

Basic Exercises

1. (1) 1.3
   (2) $-\frac{8}{17}$
   (3) 0.4
   (4) $-\frac{3}{5}$
2. C
3. ± 2
4. (1) $\frac{8}{27}$
   (2) $\frac{8}{3}$
   (3) $-\frac{8}{27}$
   (4) $\frac{8}{3}$
5. B

6. B
7. B
8. 3
9. (1) 5.62×10^{-3}
   (2) 1452
10. (a) 7.54×10^{7}; (b) 6.8×10^{-5}

Extensive Exercises

1. B
2. No, it isn't in standard form. It should be 2.44×10^{8}
3. 0.038×10^{-6}, 3.6×10^{-7}, 3800×10^{-10}, 380

Extensive Challenge

1. A
2. 0

Lesson 5 Laws of Arithmetic

Basic Exercises

1. D
2. 78
3. (1) $\frac{1}{2}$
   (2) -13
4. (1) -8
   (2) $-\frac{1}{3}$
5. (1) -10
   (2) 75
6. 0
7. $\frac{19}{2}$
8. 0
9. 11
10. 2007

Extensive Exercises

1. 6
2. -60
3. 144

Extensive Challenge

1. 1245
2. (1) $82\frac{1}{56}$
   (2) $1009\frac{1}{2018}$

Lesson 6 Prime Factors/LCM/HCF

Basic Exercises

1. (1) $420 = 2^2 \times 3 \times 5 \times 7$
   (2) $1620 = 2^2 \times 3^4 \times 5$
2. (1) 140
   (2) 2800
3. (1) 50
   (2) 1500

4. Find the HCF and LCM of each pair of numbers.

	70 and 84	52 and 78	17 and 34	7 and 8
HCF	14	26	17	1
LCM	420	156	34	56

5. There are 15 people at most. Each person gets 5 burgers.
6. 08 : 03 am
7. HCF = 12; LCM = 1200
8. HCF = 8; LCM = 45600
9. 9
10. 3

Extensive Exercises

1. £25.80
2. $p = 4, q = 2, r = 1$
3. 12, 36
   24, 36
   48, 108
   (Other reasonable answers are acceptable)

Extensive Challenge

1. 30
2. 10 : 30 am

Lesson 7 Percentages

Basic Exercises

1. C
2. 423
3. 234
4. 25
5. 1.25 kg
6. 116
7. 1.7 m
8. 25
9. 300
10. 20

Extensive Exercises

1. 40
2. (1) £26000
   (2) £39000
3. 3736

Extensive Challenge

1. D
2. 50

Lesson 8 Indices

Basic Exercises

1. 9^{10} mm^2
2. 17^3 km

3. (1) p^7
   (2) g^2
   (3) k^6
4. (a) $6x^3y^5$; (b) $6x^3y^5 9m^{14}$; (c) $27x^6y^{12}$
5. $2a^3b$
6. (1) $m = 13$
   (2) $m = 15$
   (3) $m = 5$
7. x^{10}
8. 8
9. 2^{33}
10. (1) 4^3 and $(2^3)^2$
    (2) 16 and 32

Extensive Exercises

1. 32
2. 3^{97}
3. $\dfrac{1}{512}$

Extensive Challenge

1. E
2. 2

Stage Test Answers

Stage Test（L1-L4）

1. 51
2. -7
3. 5
4. (1) -45.6
   (2) 1.5
   (3) -33.3
5. (1) 20
   (2) -25.7
   (3) -41.8
6. (1) $-5\frac{6}{7}$
   (2) $-6\frac{7}{8}$
   (3) $-1\frac{3}{20}$
   (4) $9\frac{5}{12}$
7. (1) 140
   (2) -180
   (3) -32
   (4) -180
8. $-\frac{1}{4}$
9. (1) 1.4
   (2) $-\frac{9}{16}$
   (3) 0.3
   (4) $-\frac{4}{5}$
10. 3 and -3

Stage Test (L5-L8)

1. C
2. 59
3. (1) $1\frac{1}{4}$

   (2) -13
4. (1) $2 \times 3^2 \times 5 \times 7$

   (2) $2 \times 3^2 \times 5^3$
5. (1) $2^3 \times 3 \times 11$

   (2) $2^5 \times 3^2 \times 11$
6. (1) 18

   (2) 5
7. D
8. 20%
9. 7^{12} cm^2
10. (1) $12a^4b^6$

    (2) $16n^{12}$

    (3) $8x^{12}y^9$

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